

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 5: Generator Design Tool

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Problem 1: Induced Voltage

a)

$$\omega_m = 2\pi f = \frac{\pi \cdot RPM}{30} = \frac{\pi \cdot 1500}{30} = 157.1 \frac{rad}{s} \quad (1)$$

b)

$$v_{lin} = \omega_m R = \omega_m \cdot \left(r + \frac{t_{coil}}{2}\right) = 157.1 \times \left(0.045 + \frac{0.010}{2}\right) = 7.85 \frac{m}{s} \quad (2)$$

c)

The loop length (longest part of the coil) laying perpendicular to the magnetic field lines will induce a voltage. The total length will be $2 \times$ the length of one side, giving; $2 \times l = 2 \times 0.1m = 0.2m$.

d)

$$V = NBv(2l) = 50 \cdot 0.2 \cdot 7.85 \cdot (2 \cdot 0.1) = 15.7V \quad (3)$$

e)

$$V = \frac{NBv(2l)N_s f_w}{\sqrt{2}} = \frac{50 \cdot 0.2 \cdot 7.85 \cdot (2 \cdot 0.1) \cdot 2 \cdot 0.9}{\sqrt{2}} = 20.0V \quad (4)$$

Problem 2: Resistance

a)

$$L_{wind} = \left(2 \cdot l + 2 \cdot \frac{2\pi \left(r + \frac{t_{coil}}{2}\right)}{N_s}\right) \cdot L_{corr} = \left(2 \cdot 0.1 + 2 \cdot \frac{2\pi \left(0.045 + \frac{0.01}{2}\right)}{2}\right) \cdot 1.10 = 0.566m \quad (5)$$

b)

$$L_{Cu} = L_{turn} N_{turn} N_s N_{phase} = 0.566 \cdot 50 \cdot 2 \cdot 1 = 56.6m \quad (6)$$

c)

$$R_{Cu} = \frac{\rho_{Cu}(1 + \alpha_{Cu}(T - T_0))}{A_{Cu}} = \frac{1.68 \cdot 10^{-8}(1 + 0.0039(30^\circ C - 20^\circ C))}{7.85 \cdot 10^{-7}} = 0.022 \frac{\Omega}{m} \quad (7)$$

d)

$$R_{phase} = R_{Cu} L_{stator} c_c = 0.022 \cdot 56.6 \cdot 1.10 = 1.38\Omega \quad (8)$$

Problem 3: Reactance and current

a)

$$L_s = L_{s0} \left(\frac{N_{turn}}{N_0}\right)^2 N_s = 0.0008 \cdot \left(\frac{50}{28}\right)^2 \cdot 2 = 5.10 \cdot 10^{-3}H \quad (9)$$

b)

$$\omega_e = \omega_m \frac{N_{poles}}{2} = 157.1 \frac{2}{2} = 157.1 \frac{rad}{s} \quad (10)$$

c)

$$X_s = \omega_e L_s = 157.1 \cdot 5.10 \cdot 10^{-3} = 0.801 \Omega \quad (11)$$

d)

$$|I| = \frac{V_{ind}}{\sqrt{(R_s + R_L)^2 + X_s^2}} = \frac{20}{\sqrt{(1.38 + 4.00)^2 + 0.801^2}} = 3.67 A \quad (12)$$

e)

$$V_{term} = IR_L = 3.67 \cdot 4.00 = 14.7 V \quad (13)$$

Problem 4: Power

a)

$$P_{load} = N_{phase} I^2 R_L = 1 \cdot 3.67^2 \cdot 4.00 = 54 W \quad (14)$$

b)

$$P_{loss_{Cu}} = N_{phase} I^2 R_s = 1 \cdot 3.67^2 \cdot 1.38 = 18.7 W \quad (15)$$

c)

$$P_{loss_m} = \left(\frac{RPM}{RPM_0} \right) P_{loss_0} + F_{loss\ offset} = \left(\frac{1500}{1000} \right) \cdot 75 + 5 = 117.5 W \quad (16)$$

d)

$$P_{gen} = P_{load} + P_{loss_{Cu}} + P_{loss_m} = 54 + 18.7 + 117.5 = 190.1 W \quad (17)$$

e)

$$\eta = \left(\frac{P_{out}}{P_{in}} \right) 100 = \left(\frac{54}{190.1} \right) \cdot 100 = 28.4\% \quad (18)$$

f)

$$T_{gen} = \left(\frac{P_{in}}{\omega_m} \right) = \left(\frac{190.1}{157.1} \right) = 1.211 Nm \quad (19)$$

g)

from assignment 4, the wind turbine torque came out as 1.2 Nm which gives $T_{diff} = 0.011$ with $T_{gen} > T_{aero}$. This means that at this rotational speed ω_m will the generator brake the rotors speed until the torques become balanced.